

Final Exam – Solutions

1. **Short problems.** You do not need to justify your answers in this part of the exam.

Suppose that X, Y are discrete random variables that take non-negative integer values with joint probability mass function $p(x, y)$. Express the following quantities in terms of $p(x, y)$. (Some answers might be in a form incorporating one or more sum. Make sure that the limits of the sums are properly identified. Your expressions should not include other functions besides $p(x, y)$.)

- (a) The probability mass function of X evaluated at 4.

Solution: $P(X = 4) = \sum_{l=0}^{\infty} P(X = 4, Y = l) = \sum_{l=0}^{\infty} p(4, l).$

- (b) $P(\max(X, Y) = 1)$

Solution: Since X and Y take only non-negative integer values, $\max(X, Y) = 1$ only if both X and Y are equal 1 or if one is 1 the other is 0. Thus

$$P(\max(X, Y) = 1) = p(0, 1) + p(1, 0) + p(1, 1).$$

- (c) $P(X - Y = 1)$

Solution: Since X and Y take only non-negative integer values, $X - Y = 1$ if and only if $X = k + 1$, $Y = k$ for some non-negative integer $k \geq 0$. Thus

$$P(X - Y = 1) = \sum_{k=0}^{\infty} p(k + 1, k).$$

- (d) $E[X \cdot e^Y]$

Solution: $E[X \cdot e^Y] = \sum_{k,l=0}^{\infty} k \cdot e^l \cdot p(k, l).$

- (e) The conditional probability mass function of X given $Y = 3$, evaluated at 6.

Solution: $p_{X|Y}(6|3) = \frac{p(6,3)}{P(Y=3)} = \frac{p(6,3)}{\sum_{k=0}^{\infty} p(k,3)}.$

If $P(Y = 3) = \sum_{k=0}^{\infty} p(k, 3) = 0$ then this is not defined.

- (f) $E[X|Y = 3]$ **Solution:** By definition,

$$E[X|Y = 3] = \sum_{k=0}^{\infty} k \cdot p_{X|Y}(k|3).$$

Substituting $p_{X|Y}(k|3)$ from (e), we get

$$E[X|Y = 3] = \frac{\sum_{\ell=0}^{\infty} \ell p(\ell, 3)}{\sum_{k=0}^{\infty} p(k, 3)}$$

If $P(Y = 3) = \sum_{k=0}^{\infty} p(k, 3) = 0$ then this is not defined.

2. We roll a fair die 20 times. Let X be the number of results we obtained exactly 3 times. (So if we get the following twenty numbers in some order: 1,1,1,1,2,2,2,3,3,3,3,3,4,4,4,5,6,6,6,6, then $X = 2$ because of the twos and the fours.) Find the expected value of X .

(You do not need to simplify your answer, but it should be a number, not a sum.)

Solution: Define random variables I_k , $k = 1, 2, \dots, 6$ to be 1 if the result k was obtained exactly three times, and 0 otherwise. Then

$$X = I_1 + I_2 + \dots + I_6,$$

and by linearity and exchangeability

$$E[X] = E[I_1] + E[I_2] + \dots + E[I_6] = 6E[I_1].$$

Now

$$E[I_1] = P(1 \text{ was obtained exactly 3 times}) = \binom{20}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{17},$$

thus

$$E[X] = 6 \cdot \binom{20}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{17}.$$

3. Suppose that X is a non-negative random variable.

- (a) We know that the mean of X is 100. How can you estimate $P(X \geq 150)$ using this (and only this) information?

Solution: Since X is non-negative, according to Markov's inequality, we have

$$P(X \geq 150) \leq \frac{E[X]}{150} = \frac{100}{150} = \frac{2}{3}.$$

- (b) Suppose that we also learn (in addition to its mean) that the variance of X is 400. How can you improve on your upper bound on $P(X \geq 150)$?

Solution: According to Chebyshev's inequality, we get

$$P(X \geq 150) = P(X - 100 \geq 50) \leq P(|X - 100| \geq 50) \leq \frac{\text{Var}(X)}{50^2} = \frac{400}{2500} = \frac{4}{25}.$$

- (c) Suppose that we also learn (in addition to its mean and variance) that X is actually the sum of a 100 i.i.d. random variables. How would you estimate the probability $P(X \geq 150)$ now?

Solution: We can apply CLT here. Since $\sqrt{\text{Var}(X)} = 20$, we get

$$P(X \geq 150) = P\left(\frac{X - 100}{20} \geq \frac{5}{2}\right) \approx P(Z \geq 5/2) = 1 - \Phi(2.5) = 1 - 0.9938 = 0.0062.$$

Here Z is the standard normal random variable.

Side note: if we denote the 100 iid random variables by $Y_1, \dots, Y_{100}$ then $E[Y_1] = \frac{E[X]}{100} = 1$ and $\text{Var}(Y_1) = \frac{\text{Var}(X)}{100} = 4$, but this is actually not required for the solution.

4. Suppose that we have the following information about the random variables X and Y :

$$E[X] = 1, \quad E[X^2] = 10, \quad E[Y] = 2, \quad E[Y^2] = 13, \quad \text{and} \quad \text{Cov}(3X + 2Y, 2Y - 3) = -6.$$

(a) Find the value of $E[X + 3Y]$.

Solution: Because of the linearity of expectation, we get

$$E[X + 3Y] = E[X] + 3E[Y] = 1 + 3 \cdot 2 = 7.$$

(b) Find $\text{Var}(Y)$.

$$\text{Solution: } \text{Var}(Y) = E[Y^2] - (E[Y])^2 = 13 - 4 = 9.$$

(c) What is $\text{Cov}(X, Y)$?

Solution: Since

$$\text{Cov}(3X + 2Y, 2Y - 3) = \text{Cov}(3X, 2Y) + \text{Cov}(2Y, 2Y) = 6\text{Cov}(X, Y) + 4\text{Var}(Y),$$

we can use (b) to get

$$\text{Cov}(X, Y) = \frac{-6 - 4\text{Var}(Y)}{6} = -\frac{42}{6} = -7.$$

(d) What is $\text{Corr}(X, Y)$?

Solution: Since

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 10 - 1 = 9,$$

we get from (b) and (c)

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = -\frac{7}{9}.$$

5. Let $X \sim \text{Exp}(1)$ and $Y \sim \text{Unif}(0, 1)$ be independent. What is the probability density function of the random variable $Z = X + Y$?

Make sure that you give the value of the PDF for all real numbers!

Solution: Because of independence, the PDF of $Z = X + Y$ is given by the convolution of PDF of X and PDF of Y :

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x) dx.$$

$f_X(x) \neq 0$ only if $x > 0$, and $f_Y(z - x) \neq 0$ only if $z - x \in (0, 1)$ or, equivalently, $x \in (z - 1, z)$. Thus,

$$f_Z(z) = 0,$$

if $z \leq 0$; if $z \geq 1$ we get

$$f_Z(z) = \int_{z-1}^z e^{-x} dx = e^{-(z-1)} - e^{-z},$$

and if $z \in (0, 1)$, we have

$$f_Z(z) = \int_0^z e^{-x} dx = 1 - e^{-z}.$$

So finally

$$f_Z(z) = \begin{cases} e^{-(z-1)} - e^{-z}, & z \geq 1, \\ 1 - e^{-z}, & z \in [0, 1), \\ 0, & z < 0. \end{cases}$$

6. X is a continuous random variable with probability density function given by $f_X(x) = 12x^2(1-x)$ for $x \in (0, 1)$ and 0 otherwise. The conditional distribution of Y given $X = x$ (for $0 < x < 1$) is uniform on the interval $(x, 1)$.

- (a) **(4 points)** Find the conditional probability $P(Y > 0.8|X = 0.6)$.

Solution: The conditional distribution of Y given $X = 0.6$ is uniform on $(0.6, 1)$ so the conditional probability of being larger than 0.8 is $\frac{1-0.8}{1-0.6} = \frac{1}{2}$.

- (b) **(4 points)** Find the conditional expectation $E[Y|X]$.

Solution: The conditional distribution of Y given $X = x$ is uniform on $(x, 1)$ so the conditional expectation given $X = x$ (for $0 < x < 1$) is $\frac{x+1}{2}$ (from the expected value of the uniform distribution). Hence

$$E[Y|X = x] = \frac{1+x}{2},$$

and replacing x with X gives

$$E[Y|X] = \frac{1+X}{2}.$$

- (c) **(4 points)** Find the probability density function of Y .

Solution: Note that X values from $(0, 1)$ and Y is between X and 1, so (X, Y) takes values from $\{(x, y) : 0 < x < y < 1\}$. The conditional density of Y given $X = x$ (for $0 < x < 1$) is the PDF of the uniform density on $(x, 1)$ which is $\frac{1}{1-x}$ on $(x, 1)$ and 0 outside. Hence the joint density of (X, Y) (for $0 < x < y < 1$) is

$$f(x, y) = f_{Y|X}(y|x)f_X(x) = \frac{1}{1-x}12x^2(1-x) = 12x^2,$$

with $f(x, y) = 0$ for all other values.

$$f(x, y) = \begin{cases} 12x^2, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

We can find the PDF of Y by integrating out x from the joint density. Assuming $0 < y < 1$:

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx = \int_0^y 12x^2dx = 4y^3,$$

and $f_Y(y) = 0$ otherwise.

$$f_Y(y) = \begin{cases} 4y^3, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

- (d) **(4 points)** Find the conditional probability density function of X given $Y = y$. (Make sure to specify when the conditional PDF is defined.)

Solution: This is given by $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$. Since we need $f_Y(y) > 0$, this is only defined for $0 < y < 1$. Then we get

$$f_{X|Y}(x|y) = \begin{cases} \frac{12x^2}{4y^3} = 3\frac{x^2}{y^3}, & 0 < x < y \\ 0, & \text{otherwise} \end{cases}$$

- (e) **(4 points)** Find two different single variable integrals for computing $E[Y]$ using the previous parts. (You do not need to evaluate the integrals.)

Solution: Using the PDF:

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 y \cdot 4y^3 dy.$$

Using the averaging principle and $E[Y|X = x]$:

$$E[Y] = \int_{-\infty}^{\infty} E[Y|X = x] f_X(x) dx = \int_0^1 \frac{1+x}{2} 12x^2(1-x) dx.$$

7. **(10 points)** You are given a fair die and a fair coin. You roll the die and flip the coin at the same time, and you repeat this once every minute. What is the probability that you roll your first 5 on the die exactly two minutes after seeing tails on the coin for the first time?

(You do not need to simplify your answer, but it should be a number, not a sum.)

Solution: Let A be the event in question, and denote by A_k the event that the first tail came on the k th flip of the coin and the first 5 came on the $(k+2)$ nd roll of the die. Using the independence of the die rolls and the coin flips we get

$$P(A_k) = \frac{1}{2^{k-1}} \frac{1}{2} \left(\frac{5}{6}\right)^{k+1} \frac{1}{6} = \left(\frac{5}{12}\right)^k \frac{5}{36}$$

Since A is the disjoint union of the events $A_1, A_2, \dots$ we get

$$P(A) = \sum_{k=1}^{\infty} \left(\frac{5}{12}\right)^k \frac{5}{36} = \frac{5}{36} \frac{5/12}{1 - 5/12} = \frac{25}{7 \cdot 36} = \frac{25}{252}$$